

Assignment for AE4202 – CFD for Aerospace Engineers

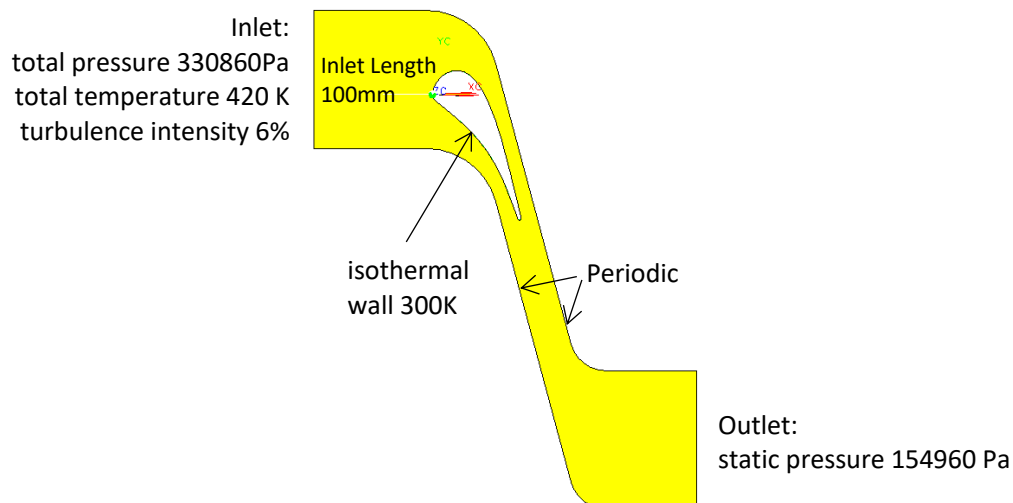
Introduction

The assignment should be performed in groups of two students. You are free to choose your partner. Both team members should equally contribute to the work and only one of them should submit the joint report. Though not recommended, you can also work alone and submit individual reports.

The deliverable is a short report (10-15 pages excluding title page and list of references) and should be submitted through Brightspace by 12 November 2026. If you submit the report late, one point will be subtracted from the grade; a second point will be subtracted if you submit more than one week past the due date.

Problem description

Your task is to perform (quasi) 2D steady state RANS simulations of a transonic nozzle guide vane. Consider the compressible flow of air, modelled as an ideal gas with a dynamic viscosity given by Sutherland's law. The geometry and boundary conditions are sketched below. Coordinates of the blade with a chord length of 67.6 mm are provided in a separate file. The computational domain is periodic with a periodicity length of 57.5mm measured in vertical direction.



The shape of the **Periodic** boundary is arbitrary; however, the lower boundary must be an exact copy of the upper boundary and shifted by exactly 57.5mm in the vertical direction. Preferably the same constraints apply to the blocking and mesh distributions. The **Inlet** boundary condition is located 100mm upstream of the blade's leading edge and should impose horizontal flow with a turbulence intensity of 6%. You may yourself choose a suitable distance for the **Outlet** boundary. Use an isothermal wall boundary condition at 300K for the guide vane.

Tasks

1. Load the profile coordinates (guide_vane.dat) and create the nozzle geometry and a suitable computational domain.
2. Mesh the computational domain with a coarse structured multi-block grid with about 20.000 cells (grid1). Pay attention to orthogonality, smoothness and resolution of the boundary layers. Visualize and describe blocking and mesh in the report and discuss the mesh quality.
3. As boundary conditions, impose an isothermal wall at the blade, total temperature, total pressure and turbulence intensity at the inflow and static pressure at the outflow. The other boundaries are periodic. Use a 'high-resolution' scheme for the spatial discretisation and the SA turbulence model. Report the simulation setup and discuss the reasons for your choices.
4. Perform a steady state simulation. Plot the solver residuals to verify convergence of the simulation.
5. Visualize the flow field: produce contour plots for the Mach number, the static pressure and temperature. State the Mach number measured near the inflow and at the outflow boundary, as well as the maximum Mach number within the full domain in a table. Plot the non-dimensional pressure coefficient C_p along the blade surface. Extract the lift and drag forces and summarize them in a second table.
6. Compute and plot the y^+ value of grid1 along the profile. Refine the grid such that $y^+ < 1$ for the refined grid (grid2). Do not exceed 400.000 cells for grid2. Visualize and describe blocking and mesh, pay attention to orthogonality and smoothness. Perform the simulation and plot the solver residuals to verify convergence of the simulation. Plot y^+ for grid2 and critically discuss the mesh quality.
7. Perform the same post processing and visualizations as done for grid1 (see point 5). Include the new C_p curve in the same figure and the Machs number and forces in the same tables as the results for grid 1 (same holds for point 8 and 9). Describe the observed flow physics and evaluate the results. Are there important differences between the grid1 and grid2 results?
8. To quantify the effect of numerical diffusion, repeat the simulations on both grid1 and grid2 with a first-order upwind scheme. Plot the solver residuals to verify convergence of the simulations. Repeat the same post processing and visualizations as done before. Analyse and discuss the effect on the results.
9. To quantify the effect of the turbulence model, repeat the simulation on the fine mesh (grid2) with a different turbulence model that can better account for anisotropic turbulence. Which model did you chose and why? Visualize and analyse the effect on the results. Also plot the solver residuals and verify convergence of the simulation.

10. Summarize your results, discuss them critically and suggest possible improvements.